

portfolio on the CAL. If you are less risk-averse than the market, then you will lever up the market portfolio and short T-bills. In this case, your portfolio will lie to the right of the market on the CAL.

Investors who are more risk-averse than Mr. Market will earn lower average returns than the market. They're perfectly fine with this because they're more risk averse, so they sacrifice the higher returns for the safety of lower-yielding T-bills. (These investors have optimal betas of less than one.) If you are more risk-seeking than Mr. Market, you will earn higher average returns than the market portfolio. You don't feel the pain of bad times as much as the average investor, enabling you to earn higher returns. (Investors more risk tolerant than Mr. Market hold portfolios with betas greater than one.) Thus, whether you have large or small holdings of market factor risk depends on whether you are more risk-averse or more risk-seeking than the market investor. In equilibrium, any person who is more risk-averse than Mr. Market is balanced by a person less risk-averse than Mr. Market.

The CAPM gives us directions on how to construct the optimal factor portfolio, which turns out to be the market-capitalization weighted portfolio of all risky assets (the market factor, of course!). It seems easy now, but when the CAPM was initially derived in the early 1960s there was no way to gain market factor exposure. It took twenty to thirty years until retail investors could gain cheap access to the market factor portfolio thanks mainly to the Vanguard Group (see chapter 17).

Value-Growth

Let's do an example of a dynamic factor allocation by taking value-growth.

There are two main theories of the value-growth premium. Rational stories define different bad times when value stocks underperform growth stocks. Some of these stories, as chapter 7 details, describe situations in which the market portfolio does badly (this is a *conditional CAPM*), so if you're already wary of bad market draws you should definitely stay away from value. We saw this happen during 2008 and 2009. But other rational stories define bad times in terms of low investment growth, or when "long run" consumption growth is low. When these episodes happen, you ask whether you can shoulder the losses in value-growth strategies more easily than the market. If so, value-growth is for you.

In the behavioral stories, value firms are neglected because investors inordinately focus on high-flying growth stocks. They overextrapolate past high growth into the future, causing them to overestimate future growth opportunities. Stocks with past high growth are bid up too high, and thus value stocks are relatively cheap. According to this line of reasoning, are you drawn to hot stocks that are in the news because they have been recently successful? If you are, you are not a good candidate for value-investing. On the other hand, if you can stomach investing in stocks out of favor or close to distress, then value investing might be your thing.